

WEP Capability Report

TASK_052 — WEP Capability Auditor and Report

WEP Capability Auditor (TASK_052)

July 5, 2026

1 Executive Summary

This report is the human-readable WEP Capability Report for TASK_052. WEP (Workflow Execution Protocol / Work Execution Protocol) is the project-local construction-loop workflow defined under `docs/dev/construction_loop/` and authorized by the main safety policy in `docs/safety.md`. The Capability Auditor inspected the repository as of HEAD 98a2040 TASK_041 add report schema checker and recorded, with evidence, what the construction loop can do today, what it cannot do yet, and what remains unproven.

The headline conclusion is conservative: the construction loop is a checker-assisted prompt-loop at most, with L2 partial automation level and no unattended runner. Loop-engineering remains probe-only and may not modify the main repo. Scientific and runtime files remain protected and are out of WEP authority.

2 Current Repo State

2.1 Worktree identity

- `pwd: /Users/wangjiahua/Desktop/25-26/Dissipation & Nonlinear Transport/newsimplification`
- `branch: task-030-loop-meta-loop-audit`
- `HEAD: 98a2040 TASK_041 add report schema checker`

2.2 Recent log

```
98a2040 TASK_041 add report schema checker
35870f5 TASK_040 add forbidden path smoke checker
c7d3b2e TASK_033 normalize report location policy
9d62546 TASK_032 backfill task history records
f900ff8 docs: add task 031 safety and reporting conventions
```

2.3 Dirty status at audit time

The four required startup commands (`pwd`, `branch`, `status`, `log`) were captured. The audit itself adds exactly the in-scope files for TASK_052 and touches no other paths.

2.4 Confirmed TASK_031 through TASK_041 state

- TASK_031 main safety and reporting conventions: landed.
- TASK_032 backfill of TASK_028 and TASK_029 history: landed.
- TASK_033 report location normalization policy: landed.

- TASK_040 forbidden path smoke checker: landed; `scripts/check_forbidden_paths.py` exists and self-tests.
- TASK_041 report schema checker: landed; `scripts/check_task_report_schema.py` exists with `--self-test` and exit codes 0/1/2.

2.5 WEP-related files found

- `docs/safety.md`
- `docs/dev/construction_loop/README.md`
- `docs/dev/construction_loop/planner.role.md`
- `docs/dev/construction_loop/executor.role.md`
- `docs/dev/construction_loop/reviewer.role.md`
- `docs/dev/construction_loop/human_integrator.role.md`
- `docs/dev/construction_loop/reporting_convention.md`
- `docs/dev/construction_loop/loop_meta_loop_repair_framework.md`
- `docs/dev/construction_loop/wep_capability_auditor.role.md` (created by TASK_052)
- `scripts/check_forbidden_paths.py`
- `scripts/check_task_report_schema.py`
- canonical task reports `reports/TASK_031_*` through `reports/TASK_052_*`

Files explicitly searched for and not found at audit time:

- `docs/dev/construction_loop/role_card_schema.md`
- `docs/dev/construction_loop/file_location_contract.md`
- `docs/dev/construction_loop/loop_engineering_merge_policy.md`
- `docs/dev/construction_loop/naming_boundary.md`
- `tasks/TASK_INDEX.md`
- `scripts/check_role_cards.py`
- `scripts/check_task_index.py`
- `scripts/run_loop_engineering_audit.py`
- `.github/`
- `agent_bus/`, `loop_engine/`, `schemas/`

3 Capability Matrix

The matrix uses the allowed status labels: IMPLEMENTED, PARTIAL, POLICY_ONLY, CHECKER_EXISTS, PROBE_ONLY, NOT_STARTED, NOT_PROVEN, DO_NOT_USE_YET.

Capability	Status	Evidence	What it can do now	What it cannot do yet	Risk
Role-card protocol	IMPLEMENTED	<code>B.role.md</code> files under <code>docs/dev/construction</code>	Defines Planner, Executor, Reviewer, HumanIntegrator	No formal schema yet; field coverage relies on convention.	Low
Planner → Executor → Reviewer → HumanIntegrator → human commit workflow	POLICY_ONLY	<code>loop_meta_loop_repair_docs/safety.md</code>	Documented lifecycle string enforced by humans.	No runner enforces the lifecycle or surfaces the next role.	Medium
Human commit gate (<code>commit now</code>)	POLICY_ONLY	<code>docs/safety.md</code> , role cards	Humans must say the exact phrase before any commit.	No machine verification that the gate was honored on every commit.	High
Safety policy	IMPLEMENTED	<code>docs/safety.md</code>	Denies forbidden paths, forbids staging, sets human gates.	Cannot enforce by itself; relies on session discipline.	Medium
Reporting convention	IMPLEMENTED	<code>docs/dev/construction</code>	None; reporting convention reports/TASK_XXX-NAME layout.	None; <code>reports/TASK_XXX-NAME</code> auto-rename legacy artifacts.	Low
human review/ PDF convention	POLICY_ONLY	<code>reporting_convention_repair framework</code>	Specifies <code>engineering_audit.tex/pdf</code> and <code>supplement/*.tex/pdf</code> .	No automation; humans compile LaTeX on demand.	Low
Engineering audit PDF standard	PARTIAL	<code>loop_meta_loop_repair</code>	Framework and allowed status words are defined.	No dedicated example, no lint, no consistency checker.	Low
Theoretical derivation supplement PDF standard	POLICY_ONLY	<code>loop_meta_loop_repair</code>	Framework defined for scientific tasks.	No actual scientific supplement exists yet in <code>reports/</code> .	Low
Task history / task index	NOT_STARTED	no <code>tasks/TASK_INDEX.md</code> found	None.	<code>tasks/TASK_INDEX.md</code> does not exist; history is reconstructed per-task.	Medium
Report location normalization	IMPLEMENTED	<code>reporting_convention.TASK_033</code>	Canonical form documented; legacy labels are frozen.	No automated migration tool for legacy artifacts.	Low
Loop-engineering merge policy	NOT_STARTED	no <code>loop_engineering_merge_policy.md</code> found	None.	TASK_034 not yet executed; decision deferred.	Medium
Loop naming boundary	NOT_STARTED	no <code>naming_boundary.md</code> found	None.	TASK_035 not yet executed.	Low
Role-card schema	NOT_STARTED	no <code>role_card_schema.md</code> found	None.	TASK_036 not yet executed; field coverage relies on convention.	Medium
Role-card backfill	NOT_STARTED	no backfill commit found	None.	TASK_037 not yet executed.	Low

Capability	Status	Evidence	What it can do now	What it cannot do yet	Risk
File-location contract	NOT_STARTED	no <code>file_location_contract.md</code> found	None.	TASK_038 not yet executed.	Medium
Task registry / index	NOT_STARTED	no <code>tasks/TASK_INDEX.md</code> found	None.	TASK_039 not yet executed.	Medium
Forbidden-path checker	CHECKER_EXISTS	<code>scripts/check_forbidden_paths.py</code> found	<code>Repaths.py</code> status <code>--short</code> , classifies against denylist, exits 0/1/2.	Cannot detect non-dirty forbidden artifacts; humans still must run it.	Low
Report checker	CHECKER_EXISTS	<code>scripts/check_task_report.py</code> found	Validates schema and optional files; recognizes <code>human_review/</code> but does not require its PDFs.	PDF/LaTeX content is not validated; humans still must compile.	Medium
Role-card checker	NOT_PROVEN	no <code>scripts/check_role_cards.py</code> found	None.	TASK_042 not yet executed.	Low
Task-index checker	NOT_PROVEN	no <code>scripts/check_task_index.py</code> found	None.	TASK_043 not yet executed.	Low
Agent-bus dry run	NOT_STARTED	no agent-bus code reachable from construction loop	None.	TASK_044 not yet executed; no agent_bus/ in main.	High
Loop-engineering audit wrapper	NOT_STARTED	no <code>scripts/run_loop_engineering_audit.py</code> found	None.	TASK_045 not yet executed.	Medium
Construction-loop runner	NOT_PROVEN	no runner exists in repo	None.	TASK_046 not yet executed; no local runner, no unattended loop.	High
GitHub Actions safety smoke	NOT_STARTED	no <code>.github/</code> found	None.	TASK_047 not yet executed; <code>.github/</code> is forbidden until a human approves it.	Medium
Langflow / cockpit readiness	NOT_STARTED	no Langflow config found	None.	TASK_051 not yet executed; Langflow integration is policy-forbidden until a human approves it.	Medium

4 What WEP Can Do Now

4.1 Human-followed protocol

WEP can be carried out by a human or single-session agent that faithfully follows the role cards. Specifically:

- Run the four startup commands and record them.
- Read `docs/safety.md` and `docs/dev/construction_loop/reporting_convention.md` and apply them.

- Write the bounded artifacts in the canonical `reports/TASK_XXX_<NAME>/` directory.
- Run the two bounded checkers (`check_forbidden_paths.py` and `check_task_report_schema.py`) on the worktree.
- Hand off to the next role via the executor report or conversational handoff.

4.2 Machine-checkable protocol

The following are machine-checkable today, on demand, by a human:

- **Forbidden dirty paths** via `python3 scripts/check_forbidden_paths.py`.
- **Report directory shape** via `python3 scripts/check_task_report_schema.py reports/TASK_XXX_<NAME>/`.

4.3 Automation not yet implemented

WEP cannot, today, automatically:

- pick the next bounded task;
- invoke Planner, Executor, Reviewer, or HumanIntegrator;
- merge or vendor loop-engineering;
- commit, push, merge, reset, rebase, cherry-pick, freeze, or sign off;
- touch `sigma_abc/`, `checkpoints/`, signoff ledgers, or scientific/runtime files.

5 What WEP Cannot Do Yet

- Cannot automatically select and safely execute the next task.
- Cannot automatically call Planner / Executor / Reviewer / HumanIntegrator.
- Cannot prove semantic or scientific correctness of any patch.
- Cannot replace the verifier, the checkpoint, or human signoff.
- Cannot let loop-engineering directly modify the main repo.
- Cannot unattended commit, push, or merge.
- Cannot safely touch `sigma_abc/`, `checkpoints/`, signoff files, or runtime files by WEP alone.
- Cannot guarantee Codex or Claude Code is in the correct worktree unless each session performs the four startup commands and records them.
- Cannot enforce the role-card schema, because the schema file does not yet exist.
- Cannot enforce the file-location contract, because the contract file does not yet exist.
- Cannot maintain the task index, because `tasks/TASK_INDEX.md` does not yet exist.

6 Automation Level Assessment

The levels used by this report are:

- **L0** — human-only docs protocol.
- **L1** — human-gated prompt-loop.
- **L2** — checker-assisted prompt-loop.
- **L3** — local semi-auto runner, no auto-commit.
- **L4** — CI-assisted runner.
- **L5** — unattended automation.

Conservative assessment: L2 partial.

Two checkers exist and can be run on demand. There is no local runner, no agent-bus integration, no GitHub Action, and no unattended loop. The role cards and reporting convention are docs-only and require humans to follow them. The audit does not rate L3 or above because no repo-local runner exists and none is authorized.

7 Loop-Engineering Status

- **Can loop-engineering be used now?**
No, probe-only.
- **Can it run in a probe worktree?**
Yes, probe-only. No probe worktree exists in the main repo.
- **Can it modify main?**
No. The denylist in `docs/safety.md` and the master repair framework forbid vendoring and submodule inclusion.
- **Can it replace WEP?**
No.
- **Can it replace verifier / signoff / checkpoint?**
No.
- **Can it be used as audit / cost helper?**
Yes, but only inside an approved wrapper, e.g. `scripts/run_loop_engineering_audit.py`, which does not yet exist (TASK_045).
- **What wrapper / checker is still needed?**
A repo-local audit wrapper that confines loop-engineering to a probe worktree, refuses forbidden-path dirt, and captures its output to a canonical `reports/TASK_XXX.<NAME>/` path.

8 Safety Boundaries for Scientific / Runtime Files

WEP alone does not authorize scientific or runtime edits.

Scientific and runtime edits remain under the verifier, the reviewer, the checkpoint, and human signoff. WEP may document, audit, and propose human-readable supplements, but it may not edit:

- `sigma_abc/`
- `checkpoints/`
- scientific output files
- validation summaries and completion matrices
- `.loop/human_signoff.yaml` and other signoff ledgers
- `agent_bus/`, `loop_engine/`, `schemas/`

Future scientific / runtime work must come through a bounded, human-approved task that explicitly names verifier, reviewer, checkpoint, and signoff constraints.

9 Recommended Next Upgrades

Only bounded next steps are recommended. The audit does not propose any automated next-step selection.

1. **Run `scripts/check_forbidden_paths.py` and `scripts/check_task_report_schema.py` as a routine pre-landing check** for every task.
2. **Resume TASK_042 role-card completeness checker** (`scripts/check_role_cards.py`) so the existing role cards can be linted automatically.
3. **Then TASK_043 task-index checker** (`scripts/check_task_index.py`) once `tasks/TASK_INDEX.md` exists.
4. **Then TASK_036 / TASK_037 / TASK_038 / TASK_039** role-card schema, role-card backfill, file-location contract, and task registry.
5. **Document the report-schema checker caveat** (`check_task_report_schema.py` requires only the four core artifacts, not the `human_review/` PDFs). A future bounded task may extend the checker to require PDFs when the task declares them; that is not in scope for TASK_052.
6. **Only after checkers mature, consider TASK_044 / TASK_045 / TASK_046** automation dry-run, audit wrapper, and local runner prototype. None of those exist today.

10 Caveats / NOT_PROVEN Items

- No WEP Capability Auditor role card existed before TASK_052. The one created here is the canonical role card; do not duplicate under another name without a bounded task.
- No `docs/dev/construction_loop/role_card_schema.md` exists at audit time. Field coverage of role cards relies on convention, not a machine-checkable schema.
- No `docs/dev/construction_loop/file_location_contract.md` exists at audit time. Path discipline relies on the role cards, the safety policy, and the reporting convention.
- The report-schema checker currently requires four core artifacts (`PLAN.md`, `executor_report.md`, `review_result.json`, `landing_report.md`) while `reporting_convention.md` and the repair framework list seven minimum files. This mismatch is recorded here as a caveat. The audit does not repair it.

- The PDF / LaTeX content quality is not validated by any checker. The auditor compiles the LaTeX on the host and records the build outcome, but does not lint the content.
- Semantic or scientific correctness is not provided by WEP. WEP does not replace the verifier, the reviewer, the checkpoint, or human signoff.
- Loop-engineering remains probe-only. This audit does not rate any loop-engineering capability as IMPLEMENTED.
- WEP is a project-local phrase. The auditor expands it in this report as Workflow Execution Protocol / Work Execution Protocol, but does not claim that WEP is an external standard.

11 Human Decision Checklist

The human reviewer is asked to confirm:

- The capability matrix is accepted as evidence-based and conservative.
- The automation level L2 **partial** is accepted.
- Loop-engineering probe-only status is accepted.
- Scientific / runtime file protection is accepted.
- The recommended next-upgrade ordering is acknowledged but not auto-scheduled.

12 Conclusion

The construction loop has a documented protocol, a canonical reporting convention, two bounded checkers, and a small number of in-flight documents (safety policy, reporting convention, repair framework). It does not have a runner, an agent bus, a GitHub Action, a Langflow integration, a role-card schema, a file-location contract, or a task index. The conservative conclusion is that WEP is an L2 partial checker-assisted prompt-loop, that loop-engineering is probe-only, and that scientific and runtime files remain protected.